COMPOSING PERFECT MORPHISMS OF SCHEMES: THE SPECTRAL SEQUENCE CAPS TOR DIMENSION AT N + M

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 17759-17762 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 17764-17780, the complete author proof (15 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Counted A side only (spaces twin proof incomplete per ruling v3).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. In-source lemma-composition-pseudo-coherent; More on Algebra, example tor (spectral sequence for composable ring maps).

Source reference. Original lemma 0681: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 17402–17405. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0662: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 17780–17780. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *A composition of perfect morphisms of schemes is perfect.*

Proof. This translates into the following algebra result: If $A \to B \to C$ are composable perfect ring maps then $A \to C$ is perfect. We have seen this is the case for pseudo-coherent in Lemma 0681 and its proof. By assumption there exist integers n, m such that B has tor dimension $\leq n$ over A and C has tor dimension $\leq m$ over B. Then for any A-module M we have

$$M \otimes^{\mathbf{L}}_A C = (M \otimes^{\mathbf{L}}_A B) \otimes^{\mathbf{L}}_B C$$

and the spectral sequence of More on Algebra, Example 0662 shows that $\mathrm{Tor}^A_p(M, C) = 0$ for $p > n + m$ as desired. □

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.